Technical Assessment Report

Problem 2: Multi-State Button Toggle and LED Flashing Frequency Control

1. Objective

The objective of this project is to develop ESP32 firmware that changes the flashing frequency of an LED through consecutive button presses. The system must use non-blocking timing (no `delay()`) to ensure button presses are registered instantly while the LED is flashing.

2. Methodology & Logic

To meet the requirements, the following logic was implemented:

- **Internal Pull-Up Resistor:** The button is connected to GPIO 14 and configured with `INPUT_PULLUP`. The idle state is HIGH, and pressing the button brings it to LOW.
- **Non-Blocking Button Debouncing:** A software timer tracks the time since the last pin state change. The system only accepts a button press if the state remains stable for at least 50 milliseconds, effectively ignoring mechanical noise (bounce).
- **State Machine (Multi-State Toggle):** A variable named `currentMode` increments on every valid button press. The modes are:
    - **Mode 0:** LED is completely OFF.
    - **Mode 1:** Slow Blink (1000 ms interval).
    - **Mode 2:** Medium Blink (300 ms interval).
    - **Mode 3:** Fast Blink (100 ms interval).
- **Concurrent Execution (`millis()`):** The `delay()` function is intentionally avoided. Instead, `millis()` is used to check the current system time against the last time the LED state was changed. If the required interval has passed, the LED toggles. This allows the main `loop()` to run continuously and catch button presses instantly.

3. Pin Configuration

- **Input:** Mode Button -> GPIO 14 (Active Low)
- **Output:** Status LED -> GPIO 25 (Active High)

4. Test Cases & Verification

Test Condition	Expected Outcome	Actual Result	Status
Initial Boot	System starts in Mode 0. LED is OFF.	LED OFF. "Mode 0: OFF" printed on Serial.	PASS
1st Button Press	System enters Mode 1. LED blinks slowly (1 sec).	LED blinks at 1000ms intervals.	PASS
2nd Button Press	System enters Mode 2. LED blinks moderately.	LED blinks at 300ms intervals.	PASS
3rd Button Press	System enters Mode 3. LED blinks very fast.	LED blinks at 100ms intervals.	PASS
4th Button Press	System loops back to Mode 0. LED turns OFF.	LED stays OFF.	PASS
Rapid/Spam Presses	Debounce logic prevents skipping modes. Changes exactly one state per press.	System advances one state cleanly per press.	PASS

5. Conclusion

The firmware successfully fulfills all requirements. By utilizing `millis()` for timing and a 50ms software debounce filter, the ESP32 accurately registers user input and concurrently updates the LED flashing frequency without any blocking delays.